Reference genome-scale metabolic models

22 models, 6 species

Species	Model	Role	Reactions	Metabolites	Genes	Exchange reactions	Biomass flux (as supplied)	Biomass flux (exchanges open)	Notes
E. coli	**iML1515**	**primary**	**2712**	**1877**	**1516**	**331**	**0.877**	**74.896**	
E. coli	**Ecoli-GEM v0.2.0 (HSDy fix)**	**primary alt.**	**2710**	**1877**	**1516**	**331**	**0.877**	**74.896**	
E. coli	Ecoli-GEM v0.2.0 (as released)	unpatched	2710	1877	1516	331	**0.000**	74.896	HSDy blocks Thr/Met/Ile synthesis
S. aureus	**iYS854**	**primary**	**1455**	**1335**	**866**	**237**	**1.360**	**391.863**	
A. baumannii	**iACB23LX**	**primary**	**2321**	**1660**	**1164**	**253**	**0.550**	**38.405**	
A. baumannii	iCN718 (BiGG)	secondary	1015	888	709	119	1.314	27.507	
A. baumannii	iCN718 (Leonidou 2024)	re-curated	1014	888	709	119	0.100	29.161	ships with minimal medium (0.1)
A. baumannii	iAB5075 (Leonidou 2024)	re-curated	2207	1525	1015	194	0.100	64.987	ships with minimal medium (0.1)
A. baumannii	iATCC1906 (Leonidou 2024)	re-curated	1270	1180	897	126	0.100	47.541	minimal medium (0.1); KEGG IDs
A. baumannii	iLP844 (Leonidou 2024)	re-curated	1628	1518	1043	171	0.100	67.601	minimal medium (0.1); ModelSEED IDs
A. baumannii	iJS784 (Leonidou 2024)	re-curated	1365	1378	969	23	**0.000**	**0.000**	cannot make biomass; ModelSEED IDs
A. baumannii	AbyMBEL891 (Leonidou 2024)	re-curated	891	916	0	8	0.100	120.371	no GPRs; 1 compartment; R### IDs
A. baumannii	AGORA AB0057 (Leonidou 2024)	re-curated	1593	1340	1019	169	0.100	133.601	minimal medium (0.1); VMH IDs
P. aeruginosa	**iSD1509**	**primary**	**2021**	**1642**	**1510**	**242**	**1.591**	**239.429**	
P. aeruginosa	iPau21	secondary	1588	1325	1169	211	**infeasible**	159.611	all exchanges closed; ModelSEED IDs
K. pneumoniae	**KpSC-pan v2**	**primary**	**3547**	**2577**	**2400**	**388**	**1.487**	**135.994**	
K. pneumoniae	iYL1228	secondary	2262	1658	1229	289	1.043	66.740	
K. pneumoniae	Jenior clinical	context	302	306	256	32	1.384	1.384	fluxes fixed by transcriptome fit; pruned
K. pneumoniae	Jenior clinical (reps)	context	384	354	328	39	2.576	2.576	fluxes fixed by transcriptome fit; pruned
K. pneumoniae	Jenior laboratory	context	290	296	253	28	0.967	0.967	fluxes fixed by transcriptome fit; pruned
K. pneumoniae	Jenior laboratory (reps)	context	393	350	330	40	2.260	2.260	fluxes fixed by transcriptome fit; pruned
E. faecium	**iDR478**	**primary**	**938**	**638**	**480**	**152**	**0.733**	**64.838**	

Biomass flux = optimal value of each model's own objective (h^{-1}). 'As supplied' uses the bounds in the published file; 'exchanges open' sets every exchange reaction to [-1000, 1000] (the objective reaction is never opened). Red = no growth. Bold rows = primary reference per species. Exchanges = cobrapy's extracellular boundary reactions (fallback: EX_ prefix). Fluxes with all exchanges open are not physiological; they show only whether the network can make biomass at all. Solver: glpk_interface, cobrapy 0.30.0.